

School of Industrial Design

Proposal

Suggestive Sketching Ai agent for enhancing user Design Ideation____
Title Title Title (Title Case)

(Thesis or Project) Keywords

Name, Name, Name (around 3)

A Project Proposal
Presented to the School of Industrial Design
Graduate Committee

By

Nicholas Economou
Master of Industrial Design Candidate 2025

In Partial Fulfillment of the
Requirements for the Degree
Master of Industrial Design in the
School of Industrial Design,
College of Architecture

Georgia Institute of Technology
April 2018

Primary Advisor

Name Name, School Name Name

Committee Members

Name Name, School Name Name

Name Name, School Name Name

Proposal

Suggestive Sketching Ai agent for enhancing user Design Ideation__

Introduction

Type your introduction including background/literature review, purpose of the project/specific aims following the “5 stages” presented in class: research problem, short literature review, deficiencies “gap” in the literature, purpose of the study/specific aims, outcomes and significance of the study. Pictures may be included if necessary representing what cannot be explained with words.

Note: the entire document MUST use Arial 10, using bold for titles only. Subtitles may use italics. The proposal is limited to minimum 6 pages and maximum 10 pages.

Introduction

Overview:

Our project aims to develop an AI-based assistive technology that enhances human sketching capabilities, specifically for product design conceptualization. By providing guidance and support through ethical decision-making processes, we aim to empower users to build their own personal datasets for refining their multimodal models while enhancing creativity and learning.

We present [SAS (Suggestive Ai Sketching)] that assists users to sketch interdependently through [scaffolding and discovery] as apposed to [following instructional guides and tutorials that are more structured and generic]. This tools is meant to support [product conceptualization to realization] with multimodal creativity support tools.

By using [suggestive feedback through predictive stroke analysis] this would in turn support human-authored product design conceptualizations, without having AI-based support taking over human leadership (*AI-in-the-Loop~\cite{stanford}* vs. *Human-in-the-loop~\cite{bigham}*).

This Project Study of [Assistive Sketching AI] provides [suggestive drawing strokes] for [design and prototyping] [for product design conceptualization], as well as [those still developing and advancing their sketching skills].

Research Problem:

We've identified the novel subject of treating AI Assistive Technologies as a pedagogical tool through personalized suggestive feedback for iterative sketching during the design process [**Citation**]. *We analyzed how to communicate to users about [their next sketching stroke], under a variety of conditions.*

The study focuses on enhancing an individual's sketching capabilities and to allow for the final outcome not directly be edited by Ai tools. The results could guide new individuals in navigating their own unique and personal approach to sketching.

We present a landscape analysis of commercially available Visual Assistance Technologies (VATs) that implement the use of recognizing and adjusting to match the users drawing style. We will also be looking into how the current market implements a pedagogical use of VATs and how they differ from our approach.

Methods:

Through literature review and keyword searches on the web and using AI models (*e.g.,[CITE]/ footnote*), we have defined a conceptual framework that is backed by prior evidence in order to scope this project outcome. Through this process, we have identified three key themes of interest: sketching and learning (1), pedagogical tools for creativity support and sketching (1), and assistive technologies and AI (1). These themes will direct the development of an AI tool that guides users in creating and refining their own sketches during the design process, without directly adding or subtracting from the final outcome. The study's results will provide valuable insights into how AI can be used as a supportive tool for human freehand drawing, thereby promoting user engagement and overall product design success

Literature Review:

Proposal

Suggestive Sketching Ai agent for enhancing user Design Ideation__

1. Introduction

Sketching is an essential and fundamental human activity for communicating ideas and bridging linguistic borders (1). Sketching can be a versatile tool for externalizing ideas for visual communication across various disciplines such as from art to design and to engineering (2). In the early stages of the design process, designers utilize freehand sketching to explore possibilities through generating ideas that can later be refined for design concepts (3). The informal/ambiguous nature of sketching enables creative thinking since it is not constrained by the precision and know-how mechanics of computer-aided design (CAD) tools (3). This means that the act of sketching is not just a motor skill but a profound cognitive activity where designers identify features and concepts for larger goals(3).

The recent addition of touchscreen devices such as digital tablets has made sketching easier due to the software applications (1). And it just so happens that there have been support and development in Artificial Intelligence (AI) (such as deep learning and generative models) that have now opened new opportunities for human-computer interactions (2). These AI systems have the potential to support and improve workflow (2). This literature review will continue to explore the three core areas relevant to the implementation of an AI-assisted sketching:

1. Sketching and Learning
2. Assistive Technologies and AI
3. Pedagogical Tools for Creativity Support and Sketching

By evaluating these areas, this literature review will identify if there are any gaps with regards to the integration of generative AI suggestions as an underlay to support (as opposed to just automating) the designer's creative process.

2. Sketching and Learning: The Cognitive role of Sketching and Constructing Mental Models

A basic understanding language of visual descriptions is when someone becomes involved the construction of mental models (4). A mental model is a cognitive representation of a real (or imaginary) perceptions formed from a person's general knowledge (4). Language for example allows one person to experience another's view of the world by dialogue. Meanwhile when someone views a sketch they construct a model based on the sketcher's design capabilities (*///Include the vosniadou paper later*). Sketches essentially are visual descriptions and are representations of these internal models (4). They are powerful (for good or bad) because they are often indeterminate meaning that they can consist of many possible interpretations based on the viewer (4).

This cognitive process is actually central to design ideation! When a designer sketches, they begin to externalize their internal mental representations and insight. This is a process that can help overcome idea fixation (3). The sketch itself acts as part of a visual dialogue where modifications on the page reflect an evolving internal mental model (3). This process is not about creating a perfect, final image but about exploring a conceptual space (3). For example, sketches that are done in the early design stages are used for exploring possibilities and generating novel ideas (3).

However sketching skills can be a barrier in it of themselves(3). The cognitive load of mastering technical sketching skills (perspective, line weights, dimensional accuracy etc) can be a threshold on the creative exploration that sketching is meant to facilitate (3). Therefore, a tool that supports the user's sketching process without taking over could be beneficial. Improving or even supporting the skill lowers the barrier to visual expression thus allowing the designer to focus more on high-level cognitive tasks of designing such as ideation (3).

3. Assistive Technologies and AI: AI based on Free-Hand Sketches and Generation

Recent advancements in AI development have significantly impacted the field of Design such as inspiration for ideation via visual sketch generations (1). Free-hand sketches present unique challenges for AI models due to their nature and ambiguity as apposed to photos and images (1). Various deep learning architectures have been developed to address these challenges by processing

Proposal

Suggestive Sketching AI agent for enhancing user Design Ideation

sketches in different representations: ranging from raster images (CNNs), stroke sequences (RNNs), and topological graphs (GNNs) (1).

3.1 Current Sketch Generation Models

The development of sketch generation such as SketchRNN, is a sequence-to-sequence variational auto-encoder (VAE) that then models the drawing process via generating vector sketches stroke by stroke (1). This approach is significant because it moved beyond treating sketches as static images and instead captured the dynamic nature of how humans draw. However, such data-driven models are often limited due to the styles within their training data. Some models can contain around 345 object categories such as in the QuickDraw dataset (2).

More recent methods utilize large-scale pre-trained vision-language models (for example: CLIP) as well as text-to-image diffusion models in order to generate sketches of more diverse concepts without the need to rely on sketch-specific datasets (2). These models often optimize in order to prioritize text prompts, however they lose the sequential step-by-step process similar to naturally drawing (2).

3.2 Current AI Sketching Tools and their Interactions

The integration of AI within creative workflows is a growing field. Intelligent tutoring systems have been developed to provide personalized feedback and guided practice to help users improve their technical sketching skills (*reference Sketchetivity in this section*). These systems use sketch recognition algorithms to interpret student drawings and provide feedback on aspects like accuracy and perspective (*reference Sketchetivity in this section*).

AI can now be utilized and explored as a collaborative partner within the creative design process. The SketchAgent project depicts and demonstrates that a large language model (LLM) can be prompted to generate sketches sequentially in a way that can collaborate with human users (2). This is achieved without any model training, by defining an intuitive "sketching language" that the LLM uses to specify strokes on a grid-based canvas. This work highlights a critical shift from AI as a tool for generating a final product to AI as an interactive agent participating in the process of creation (2). This concept of human-AI co-creation is what started this projects initiative and the core for how to improve on its implementation in the design field.

4. Pedagogical Tools: Human-AI Collaboration in Design and Learning

The introduction of Generative Artificial Intelligence (GAI for short) tools within design education has the potential to reshape traditional learning methods such as with Bloom's Taxonomy (5). A case study involved students who were tasked with designing a robot and they had access to GAI tools. This study found that the AI's role within the problem solving process changed depending on the students' familiarity with the task. When students were knowledgeable (with sketching or CAD), the AI tool served as an accelerator that enhanced their ability to analyze and evaluate and problem solve (5). However, when faced with an unfamiliar domain (for example: Arduino programming), students used GAI tools to directly create solutions before they fully understood the underlying problem (5). This suggests that Generative AI can enable incorrect learning where practical application to problem solve precedes theoretical understanding of the problem(5).

This can be a valuable reason for designing an AI-assisted sketching tool. The AI tool should act as a scaffold that enables users to explore complex design variations they might not have the technical skill to produce on their own (5). With the robot design project example in mind, GAI tools were used for brainstorming initial concepts as well as adding depth to visual representations and renderings (5). However obtaining those skills to begin with requires critical thinking and judgment from the user as that is how they can evaluate the AI's output (5). Basically students' ability to be critical when evaluating GAI outputs varied based on their prior knowledge and understanding of what they were solving for.

This again highlights the core motivation for this project: creating a tool where the AI provides suggestions as an *underlay*. This is because this interaction can keep the human designer firmly in control of the design process all while using the AI's output as inspiration or as a suggestion rather

Proposal

Suggestive Sketching Ai agent for enhancing user Design Ideation__

than asking for a final solution. It also stays relevant to the traditional methods such as using tracing paper when creating visual design and needing multiple layers for more details. The user can retain more authorship as they are the one making conscious decisions about which AI suggestions to incorporate, or to ignore altogether. This non-intrusive approach could potentially foster a healthier and more ethical collaborative and even educational partnership that can utilize new technologies while at the same time preserving creative ownership.

Gaps in Literature:

The literature reveals significant progress in understanding the cognitive understanding of sketching for exploring human-AI collaboration in creative tasks. However a gap exists at the intersection of creativity support and pedagogy. While tools like SketchAgent focus on interactive co-design application, and Sketchivity that applies digital itself as an educational tool, there is less focus on a specific interaction where AI can be used as a non-intrusive support agent.

Much of the research on AI sketching tools is either software-based, or focused with a hardware platform. My project addresses this by creating an integrated system utilizing commercially available microprocessors (Raspberry Pi). This approach allows for a more holistic design where the hardware and software are specifically created for AI-assisted sketching design workflows.

Purpose and Aims:

Outcomes and Significance:

School of Industrial Design

Proposal

Suggestive Sketching Ai agent for enhancing user Design Ideation__

Methodology

Type your methodology here including methods, design process, participants, materials, procedures, etc answering methodology questions presented in class.

School of Industrial Design

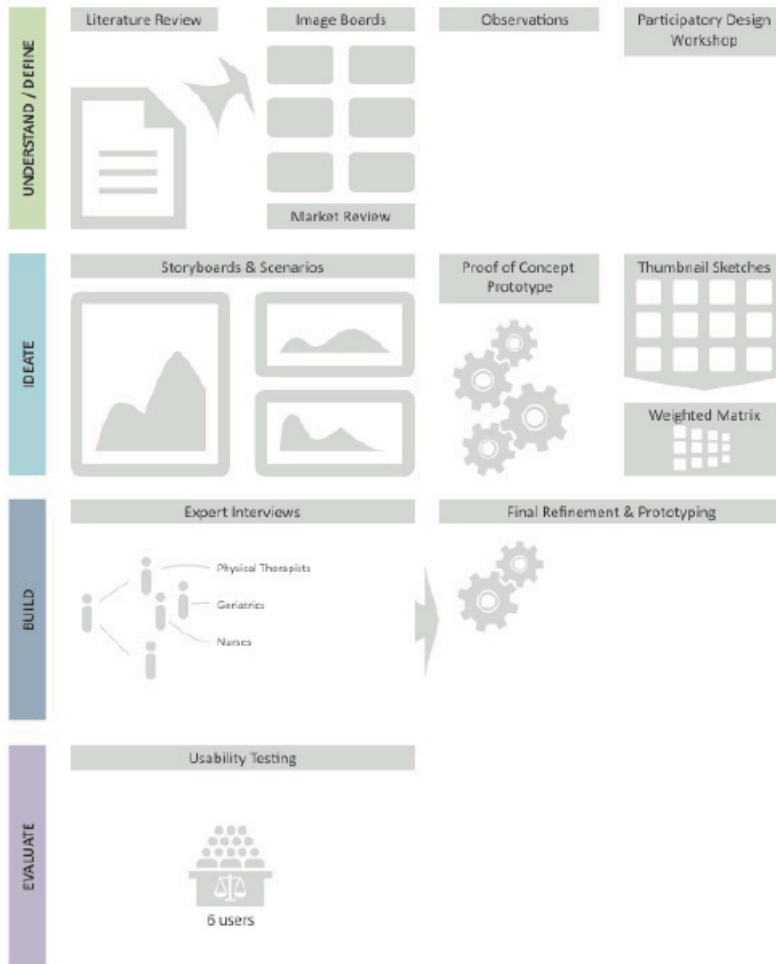
Proposal

Suggestive Sketching Ai agent for enhancing user Design Ideation__

Methodological Map

Insert here a visual map of the thesis/project methodology.

Example:



School of Industrial Design

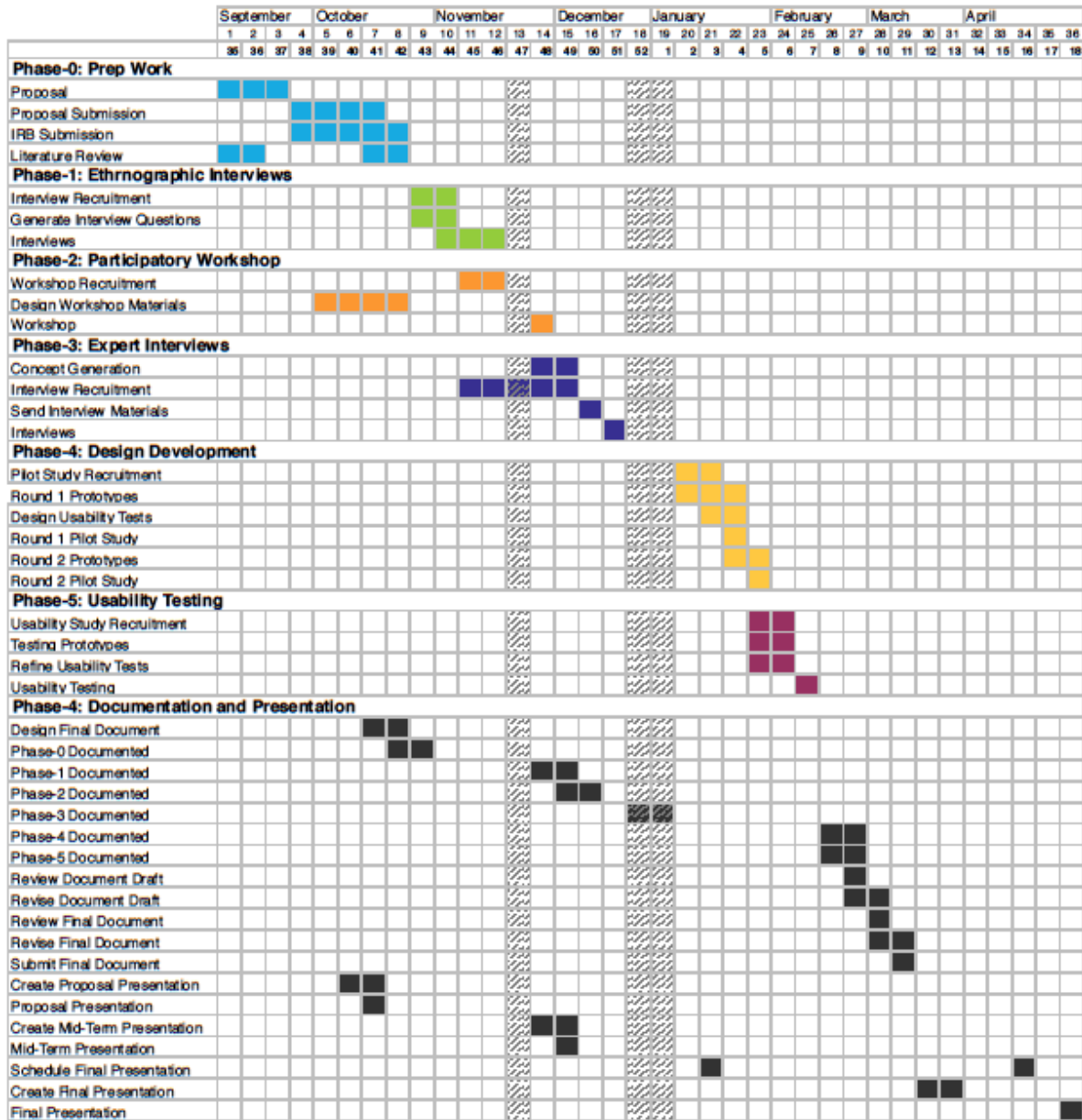
Proposal

Suggestive Sketching Ai agent for enhancing user Design Ideation__

Timeline

Insert here a visual timeline of your thesis/project.

Example:



Proposal

Suggestive Sketching Ai agent for enhancing user Design Ideation__

References

List all references used in this proposal following the APA 6th style (preferably).

- (1) Xu, P., Hospedales, T. M., Yin, Q., Song, Y.-Z., Xiang, T., & Wang, L. (2023). Deep Learning for Free-Hand Sketch: A Survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(1), 285–312. <https://doi.org/10.1109/tpami.2022.3148853>
- (2) Vinker, Y., Shaham, T. R., Zheng, K., Zhao, A., Fan, J. E., & Torralba, A. (2024). SketchAgent: Language-Driven Sequential Sketch Generation. *ArXiv (Cornell University)*. <https://doi.org/10.48550/arxiv.2411.17673>
- (3) Merzdorf, H. E., Jaison, D., Weaver, M. B., Linsey, J., Hammond, T., & Douglas, K. A. (2023). Sketching assessment in engineering education: A systematic literature review. *Journal of Engineering Education*. <https://doi.org/10.1002/jee.20560>
- (4) Comprehension as the construction of mental models. (1981). *Philosophical Transactions of the Royal Society of London. B, Biological Sciences*, 295(1077), 353–374. <https://doi.org/10.1098/rstb.1981.0145>
- (5) Jiménez Romanillos, Elena & Andersson, Torbjörn. (2024). THE INTERSECTION OF AI AND BLOOM'S TAXONOMY IN DESIGN EDUCATION: A ROBOTIC DESIGN CASE STUDY. 651-656. 10.35199/EPDE.2024.110.

Examples:

Book:

Author, A. A. (Year of publication). *Title of work: Capital letter also for subtitle*. Location: Publisher.

Journal Articles/Periodicals:

Author, A. A., Author, B. B., & Author, C. C. (Year). Title of article. *Title of Periodical*, volume number(issue number), pages. doi:http://dx.doi.org/xx.xxx/yyyy

Website:

Author, A. A., & Author, B. B. (Date of publication). *Title of document*. Retrieved from http://Web address

Citation details:

<http://owl.english.purdue.edu/owl/resource/560/1/>